AI-Powered-Healthcare-Diagnosis-Assistant

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Abstract

This project is an AI-Powered Healthcare Diagnosis Assistant that predicts diseases based on the symptoms selected by the user. The system uses Machine Learning algorithms to analyze symptoms and provide disease predictions. It also displays recommended medicines, diet plans, precautions, and the type of doctor to consult. The application is developed using Python, Streamlit, Pandas, Scikit-learn, and Pickle. This project is created for educational purposes to demonstrate the use of Artificial Intelligence in healthcare.

INTRODUCTION

Healthcare is one of the most important fields where Artificial Intelligence is making a significant impact. Many diseases can be identified at an early stage by analyzing symptoms with Machine Learning techniques. This project, AI-Powered Healthcare Diagnosis Assistant, is developed to predict diseases based on user-selected symptoms. It provides information such as recommended medicines, diet plans, precautions, and the type of doctor to consult. The application is built using Python, Streamlit, Pandas, Scikit-learn, and Pickle. The main objective of this project is to provide a simple, fast, and user-friendly healthcare assistance system for educational purposes.

Objectives

- To predict diseases based on user-selected symptoms.
- To provide quick and accurate disease prediction using Machine Learning.
- To recommend medicines, diet plans, and precautions.
- To suggest the appropriate doctor for consultation.
- To create a simple and user-friendly healthcare application.
- To demonstrate the practical use of Artificial Intelligence in healthcare.

Technologies Used

Technology	Purpose
Python	Programming Language
Streamlit	Web Application Framework
Pandas	Data Processing
Scikit-learn	Machine Learning Model
Pickle	Saving and Loading Model
HTML	Basic Web Structure
CSS	User Interface Styling
Git	Version Control
GitHub	Project Repository
VS Code	Code Editor

System Requirements

Requirement	Specification
Operating System	Windows 10 / Windows 11
Programming Language	Python 3.x
IDE	Visual Studio Code
Framework	Streamlit
Libraries	Pandas, NumPy, Scikit-learn, Pickle
Browser	Google Chrome / Microsoft Edge
RAM	Minimum 4 GB
Storage	Minimum 500 MB Free Space

Project Workflow

1. User opens the AI Healthcare Diagnosis Assistant.
2. User selects symptoms from the available list.
3. The selected symptoms are converted into machine-readable input.
4. The trained Machine Learning model predicts the disease.
5. The application displays:
    a. Predicted Disease
    b. Recommended Medicine
    c. Recommended Diet
    d. Precautions
    e. Doctor to Consult
6. The user reviews the results and consults a doctor if necessary.

Workflow Diagram

Start

↓

Select Symptoms

↓

Machine Learning Model

↓

Disease Prediction

↓

Medicine + Diet + Precautions + Doctor

↓

End

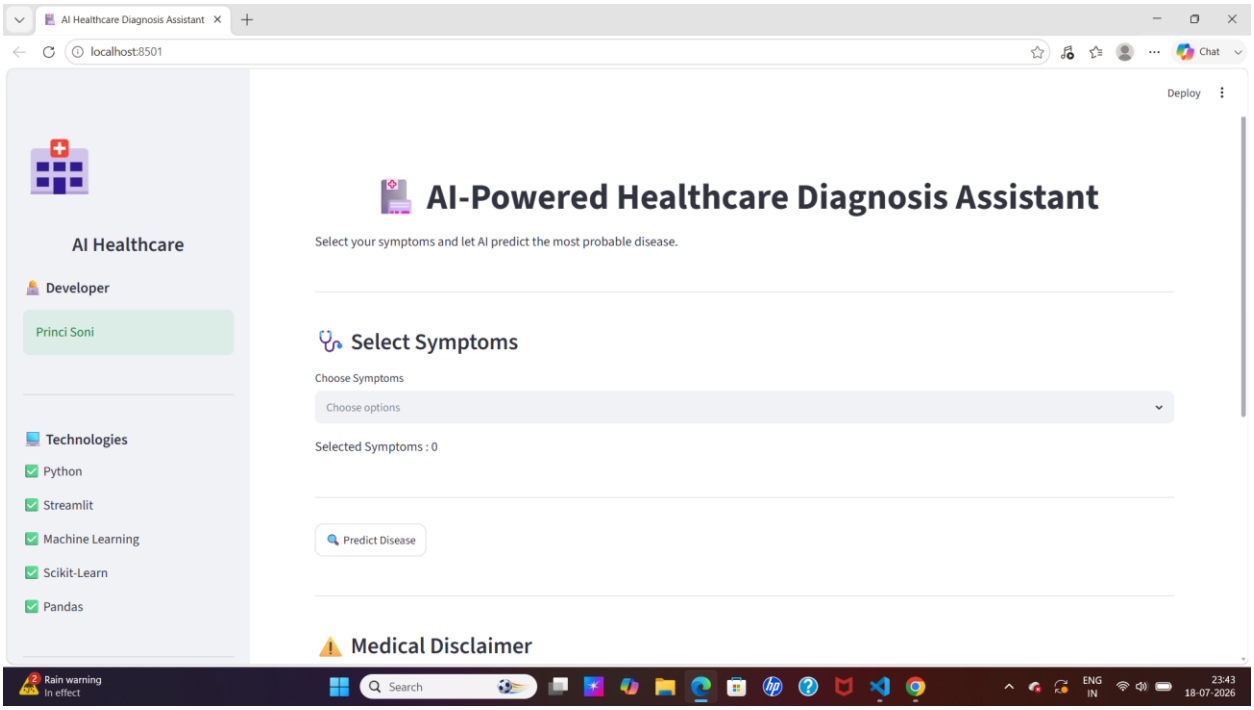
Machine Learning Model

The AI-Powered Healthcare Diagnosis Assistant uses a Machine Learning model to predict diseases based on the symptoms selected by the user. The model was trained using a healthcare dataset containing various symptoms and their corresponding diseases. During prediction, the selected symptoms are converted into binary values (0 and 1), which are passed to the trained model. The model analyzes the input and predicts the most probable disease. The predicted disease is then used to display medicines, diet recommendations, precautions, and the type of doctor to consult.

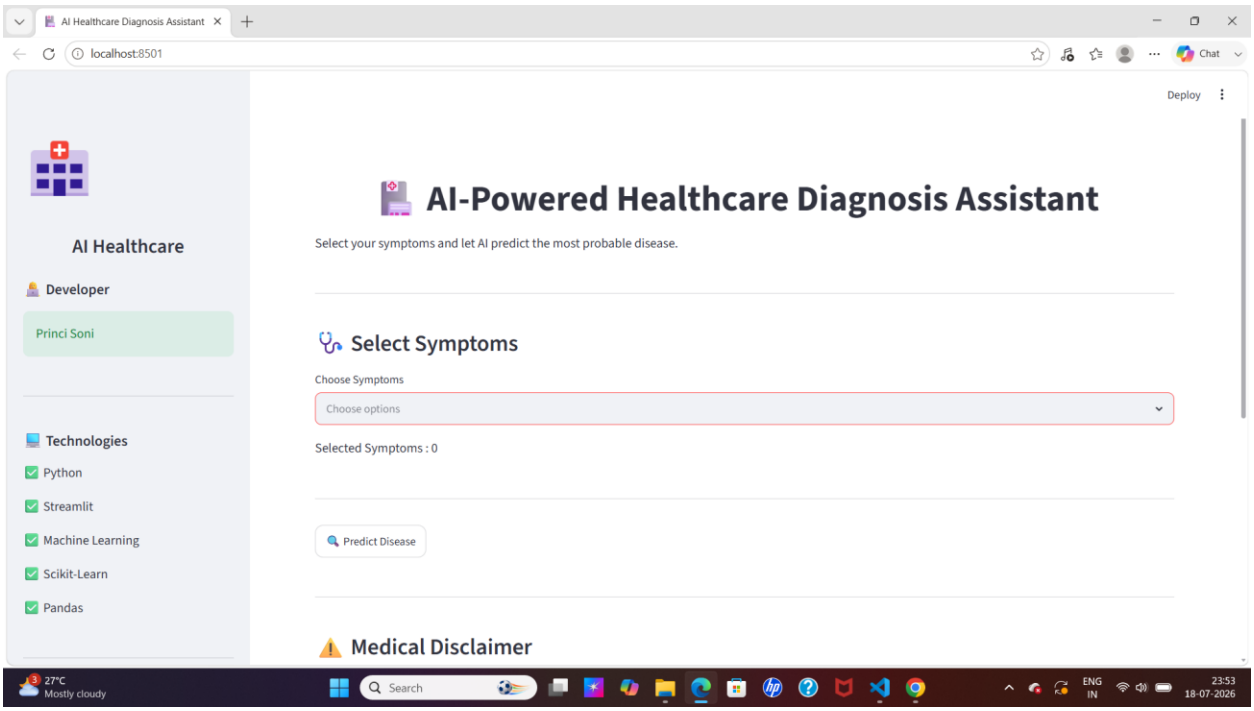
Machine learning Processes

- Data Collection
- Data Preprocessing
- Model Training
- Model Saving using Pickle
- Disease Prediction
- Displaying Results

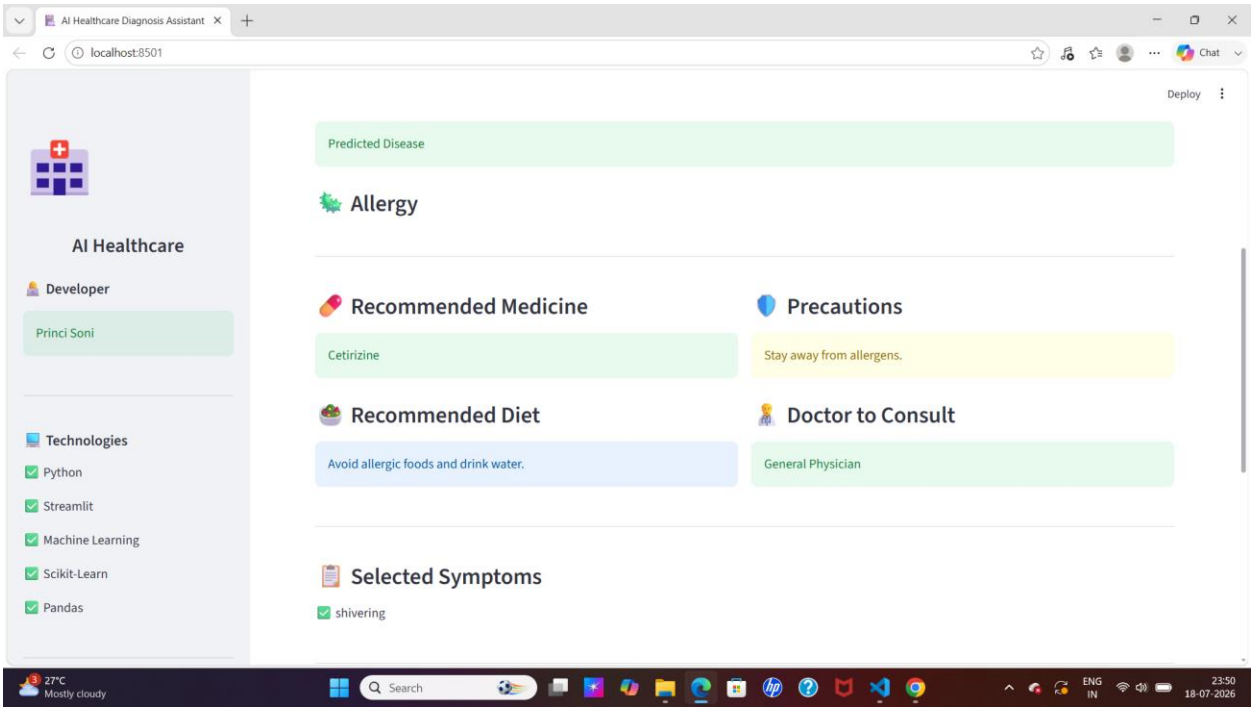
Project Output



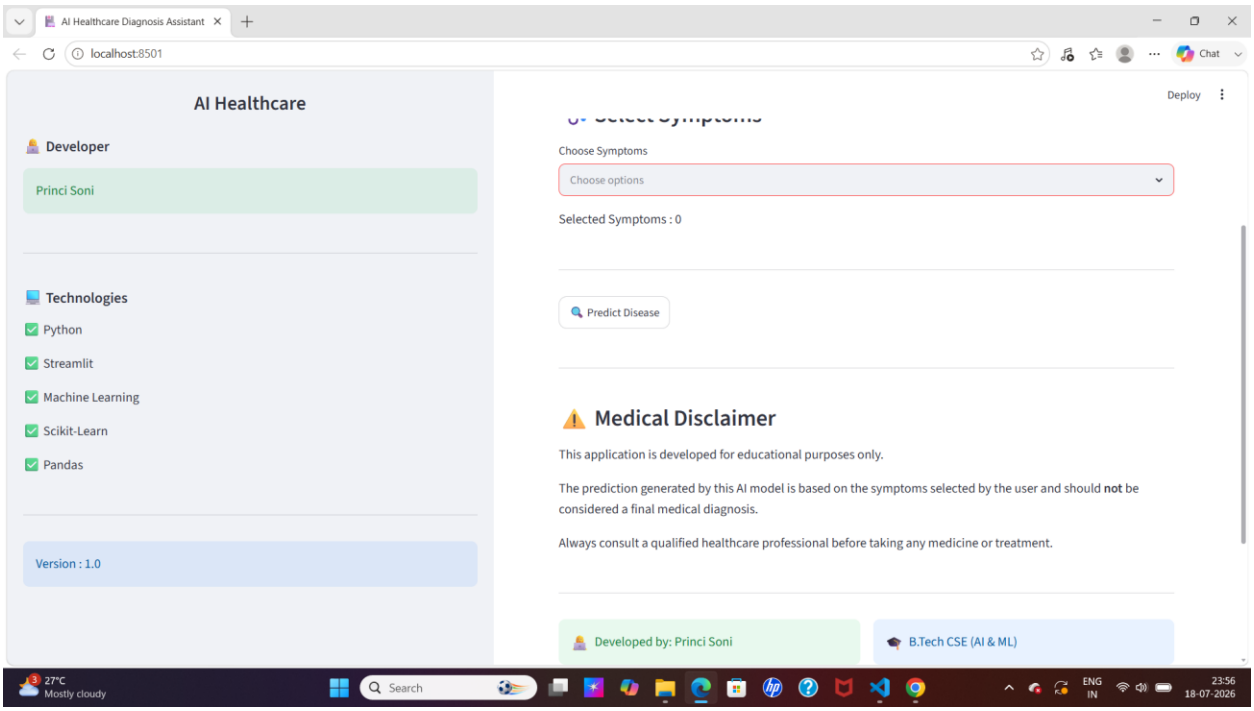
Home Page



Symptoms Selection



Disease Prediction Result



Sidebar/Developer Details

CONCLUSION:

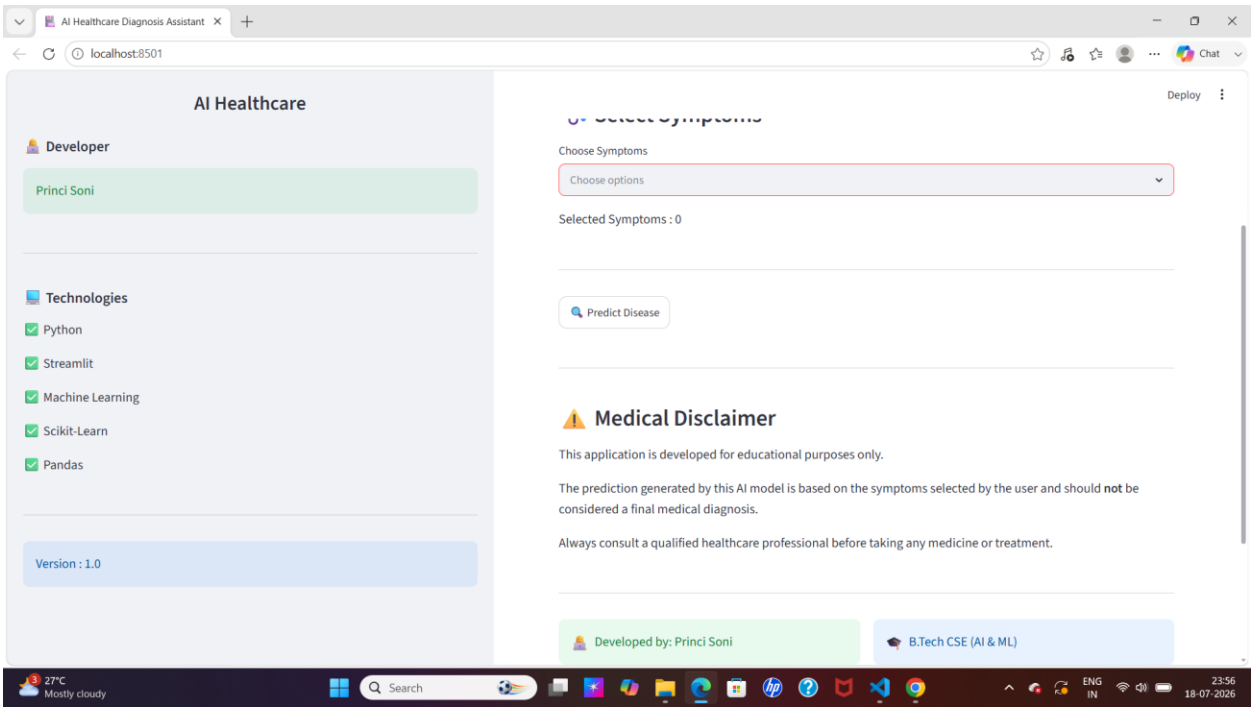
The AI-Powered Healthcare Diagnosis Assistant is a useful Machine Learning application that predicts diseases based on user-selected symptoms. It provides medicine recommendations, diet suggestions, precautions, and doctor consultation guidance in a simple and user-friendly interface. This project demonstrates how Artificial Intelligence and Machine Learning can be applied in the healthcare domain. Although it is developed for educational purposes, it shows the potential of AI to assist users in obtaining quick health-related information.

Future Scope

The project can be further improved by integrating real-time hospital and doctor databases. Future versions may include chatbot support, voice-based symptom input, online appointment booking, medical report analysis, and medicine reminders. Deep Learning models and larger healthcare datasets can also be used to improve the accuracy of disease prediction. A mobile application version can make the system more accessible to users.

Reference

- Python Official Documentation – https://docs.python.org/3/
- Streamlit Documentation – https://docs.streamlit.io/
- Scikit-learn Documentation – https://scikit-learn.org/
- Pandas Documentation – https://pandas.pydata.org/
- NumPy Documentation – https://numpy.org/
- GitHub – https://github.com/


AI Healthcare Diagnosis Assistant
localhost:8501
Chat
AI Healthcare
Developer
Princi Soni
Technologies
Python
Streamlit
Machine Learning
Scikit-Learn
Pandas
Version : 1.0
Deploy
Choose Symptoms
Choose options
Selected Symptoms : 0
Predict Disease
Medical Disclaimer
This application is developed for educational purposes only.
The prediction generated by this AI model is based on the symptoms selected by the user and should not be considered a final medical diagnosis.
Always consult a qualified healthcare professional before taking any medicine or treatment.
Developed by: Princi Soni
B.Tech CSE (AI & ML)
27°C
Mostly cloudy
Search
ENG
IN
23:56
18-07-2026